Example 4.25 PQ is a chord of length 8 cm to a circle of radius 5 cm. The tangents at P and Q intersect at a point T. Find the length of the tangent TP.

Solution Let $TR = y$. Since, OT is perpendicular bisector of PQ.

$PR = QR = 4$ cm

In ΔORP, $OP^2 = OR^2 + PR^2$

$OR^2 = OP^2 - PR^2$

$OR^2 = 5^2 - 4^2 = 25 - 16 = 9 \Rightarrow OR = 3$ cm

$OT = OR + RT = 3 + y$ ... (1)

In ΔPRT, $TP^2 = TR^2 + PR^2$ ... (2)

and ΔOPT we have, $OT^2 = TP^2 + OP^2$

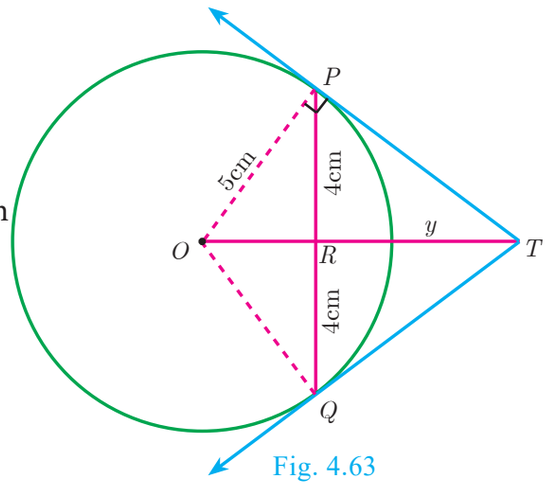


Fig. 4.63

$OT^2 = (TR^2 + PR^2) + OP^2$ (substitute for TP^2 from (2))

$(3 + y)^2 = y^2 + 4^2 + 5^2$ (substitute for OT from (1))

$9 + 6y + y^2 = y^2 + 16 + 25$

$6y = 41 - 9$ we get $y = \frac{16}{3}$

From (2), $TP^2 = TR^2 + PR^2$

$TP^2 = \left(\frac{16}{3}\right)^2 + 4^2 = \frac{256}{9} + 16 = \frac{400}{9}$ so, $TP = \frac{20}{3}$ cm

Example 4.26 In Fig.4.64, O is the centre of a circle. PQ is a chord and the tangent PR at P makes an angle of 50° with PQ. Find $\angle POQ$.

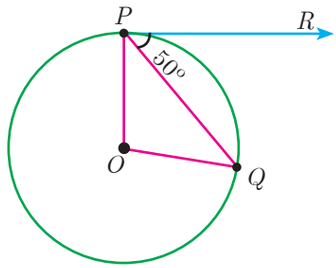


Fig. 4.64

Solution $\angle OPQ = 90° - 50° = 40°$ (angle between the radius and tangent is 90°)

$OP = OQ$ (Radii of a circle are equal)

$\angle OPQ = \angle OQP = 40°$ ($\triangle OPQ$ is isosceles)

$\angle POQ = 180° - \angle OPQ - \angle OQP$

$\angle POQ = 180° - 40° - 40° = 100°$

Example 4.27 In Fig.4.65, ΔABC is circumscribing a circle. Find the length of BC.

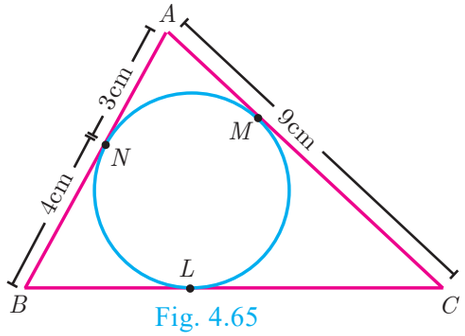


Fig. 4.65

Solution $AN = AM = 3$ cm (Tangents drawn from same external point are equal)

$BN = BL = 4$ cm

$CL = CM = AC - AM = 9 - 3 = 6$ cm

Gives $BC = BL + CL = 4 + 6 = 10$ cm

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